Multilevel Denoising for High-Quality SAR Object Detection in Complex Scenes

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***Abstract*— Deep learning-based methods have dominated in object detection in synthetic aperture radar (SAR) images. Despite significant advancements, existing approaches mainly focus on architectural enhancements of the network, leaving the unique challenges posed by the strong speckle noise inherent in SAR imagery not fully tackled. In this article, we introduce the multilevel denoising detection transformer (MD-DETR) to mitigate the impact of speckle noise for SAR object detection. We build MD-DETR in three steps, leveraging the advanced real-time detection transformer (RT-DETR) framework. First, we address the noise data input by designing a straightforward image-level denoising technique, which concatenates the original image with its denoised counterparts to create a new image for model input. Subsequently, for feature-level denoising, a coarse-mask guidance feature learning module is proposed to enhance the global features of the decoder. In addition, for query-level denoising, we introduce an auxiliary head with the one-to-many matching strategy. This enables flexible query selection by dynamically adjusting the number of queries to suit various scenarios. To validate the superiority of the proposed MD-DETR, extensive experiments are conducted on two benchmark datasets, that is, the SAR ship detection dataset (SSDD) and the recently published COCO-level large-scale multiclass SAR object detection dataset (SARDet-100K). Experimental results on both datasets outperform previous advanced detectors, achieving a new state-of-the-art (SOTA) with 98.9 AP_{50} and 88.5 mAP_{50} on SSDD and SARDet-100K, respectively.**

***Index Terms*— Detection transformer (DETR), multilevel denoising, object detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) imaging technology plays a pivotal role in geospatial intelligence due to its irreplaceable ability to operate under various weather conditions and at any time of the day [1]. Despite the advancements in SAR imaging, the detection of multiclass objects in SAR images remains a challenging task, primarily due to the strong speckle noise and background clutter.

Although deep learning-based methods have demonstrated remarkable success in object detection within SAR imagery, existing approaches primarily focus on architectural enhancements of the network, neglecting the unique challenges posed by the strong speckle noise inherent in SAR imagery, which has not been fully addressed.

In general object detection, existing methods generally adopt the CNN-based architecture, the most famous of which are among the faster R-CNN series [2], [3], [4] and the YOLO series [5], [6]. Nevertheless, all the above typically require nonmaximum suppression (NMS) for both training and inference, which inevitably introduces hyperparameters for tuning. Bypassing the NMS operation, the end-to-end Transformer-based detectors (DETRs) [7], [8], [9], [10], [11], [12] have received extensive attention from the academia due to their streamlined pipeline. However, the high computational demands of these methods hinder real-time detection, rendering the DETR framework without a distinct advantage in terms of inference speed. To tackle this issue, a real-time detection Transformer (RT-DETR) [13] is proposed to achieve an ideal tradeoff between speed and accuracy.

Though the aforementioned methods have made significant progress in general object detection, the noise issue in SAR imagery is not adequately addressed. Speckle noise, an artifact resulting from the coherent processing of SAR data, presents a unique challenge for object detection because of its non-Gaussian and multiplicative nature. This type of noise is not merely additive like in traditional imaging systems but arises from the coherent reflection of radar signals from the scene's surface. Thus, standard denoising techniques on the image-level are inadequate to tackle this challenge. Moreover, the granular texture introduced by speckle noise can obscure fine details and significantly reduce the signal-to-noise ratio, making it difficult to differentiate objects from the background at the feature level. Thus, denoising at the feature level becomes essential to enhance the discriminative capabilities of the detector and preserve the semantic information. Besides, the speckle noise can cause the detection algorithm to mistakenly identify noise patterns as part of the object, leading to the initialization of bounding boxes that are either false positives or inaccurately located. Therefore, denoising at the initial query level is also important to ensure that the detector starts with a cleaner and more accurate representation of potential objects. In summary, the speckle noise requires a different approach that can handle its multiplicative effect on the image intensity.

However, the application of denoising strategies to SAR object detection has not been extensively explored. To this end, in this article, we pay attention to the noise issue at the level of data input, encoder feature, and initial query

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selection. SAR imagery's unique attributes demand a sophisticated denoising strategy, thus our approach is designed to systematically address the noise at the pixel, feature, and query levels. Image-level denoising is the first step to reduce speckle noise at the pixel level. Feature-level denoising is implemented through a novel feature learning module. Finally, query-level denoising addresses the challenge of varying object densities in SAR images. Specifically, we build upon the RT-DETR framework and combine multilevel denoising mechanisms, tailored for SAR object detection. The reason we chose RT-DETR is due to its reasonable balance between speed and accuracy. In the following, we will introduce the proposed denoising strategies.

First, from the perspective of the data input, SAR images are inherently single-channel. However, to accommodate pretrained models from ImageNet, existing SAR detectors [14], [15], [16] often simply replicate the single-channel SAR image to create a three-channel RGB-like image. This straightforward replication, though, does not yield any additional information gain, making it a suboptimal approach. Thus, we first propose an image-level denoising approach that leverages traditional image-denoising algorithms, such as Gaussian filtering and bilateral filtering, to perform denoising at the pixel level. Subsequently, the denoised image is concatenated with the original image along the channel dimension, forming a genuine three-channel image that serves as input to the detection model. Notably, the filtered image, with its reduced noise intensity, complements the original image effectively, potentially enhancing the feature representation for the detection task.

Second, from the perspective of the denoised feature representation, although the original RT-DETR uses a hybrid encoder to produce multiscale features for the decoder, it does not explicitly perform denoising at the feature level, as natural images do not exhibit the speckle noise characteristic of SAR imagery. This inevitably results in encoder features with low discriminative and localization confidence, leading to a considerable level of uncertainty and hurting the detection performance. Different from existing detectors, we do not engage in explicit feature-level denoising. Instead, we adopt implicit feature-level denoising in a weakly supervised manner. Specifically, a coarse-mask guidance feature learning module is proposed to bolster the encoder's representation capability, ensuring a more discriminative and robust feature for the decoder.

Third, we view denoised query initialization as a breakthrough to further improve performance. The original RT-DETR proposes the uncertainty-minimal query selection strategy, which, despite its high quality, necessitates a fixed number (e.g., 300) of queries per image. Given that the number of objects in each SAR image can vary significantly, the fixed quota of initial queries leads to unnecessary computational overhead. For example, the static allocation of a large number of queries results in excessive computation for images with fewer objects. To address this inefficiency, we propose a query-level denoising strategy on dynamically selecting a smaller, yet more accurately localized initial query. We observed that during the initial training phase, the one-to-many matching strategy in anchor-based detectors yielded higher quality positive samples more readily compared to the one-to-one matching adopted in query-based detectors. This motivates us to introduce an auxiliary head with the one-to-many matching strategy to support flexible query selection, in which the number of queries is dynamically adjusted based on the object density within an image. In this way, the computational load is reduced while the accuracy is even improved.

The above three denoising strategies constitute our proposed multilevel denoising detection Transformer (MD-DETR). More specifically, the main contributions of our work are summarized as follows.

1) *Image-Level Denoising for Data Input*: We introduce an innovative image-level denoising approach that employs traditional denoising algorithms such as Gaussian and bilateral filtering to process single-channel SAR imagery. By fusing the denoised image with the original along the channel dimension, we construct a three-channel input for the detection model without the redundancy of simple replication.
2) *Feature-Level Denoising for Enhanced Encoder Features*: We propose a weakly supervised implicit feature-level denoising strategy. A coarse-mask guidance feature learning module is integrated to fortify the encoder's feature representation capabilities, leading to more discriminative and robust features for the decoder, which are crucial for accurate SAR object detection.
3) *Query-Level Denoising for Accurate Query Initialization*: We present a query-level denoising strategy that dynamically adjusts the number of initial queries based on the object density in the SAR image. This approach eliminates the fixed quota of the query number, thereby reducing the redundant computational overhead triggered by the cross-attention between queries and encoded features in the decoder.

To validate the effectiveness of MD-DETR, extensive experiments are carried out on two benchmark datasets, that is, the SAR ship detection dataset (SSDD) [17] and the recently published COCO-level large-scale multiclass SAR object detection dataset (SARDet-100K) [1], and our method achieves competitive results compared to current SOTAs.

The rest of this article is organized as follows. Section II gives the related work. Section III introduces the proposed MD-DETR in detail. The experimental results and analysis are presented in Section IV. Section V gives the conclusion and future work.

II. Related Work

A. Deep Learning-Based General Object Detection

In the area of object detection, deep learning has achieved landmark success due to its rich feature representation. The current deep learning-based object detectors can be classified into two types based on their detection heads: CNN decoder-based detectors and Transformer decoder-based detectors.

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CNN decoder-based detectors typically start by obtaining the default anchor-boxes or anchor-points from the original images, then classify and refine these anchors into accurate bounding boxes. Most two-stage detectors, such as faster R-CNN [2] and cascade R-CNN [3], generate anchors in the region proposal network (RPN) and progressively classify and refine them in downstream classifiers, though with high accuracy, the speed is unsatisfactory in real-time applications. Toward real-time object detection, starting from YOLOv1 [18], the YOLO series [5], [6], [19], [20], [21], [22] have outperformed other one-stage object detectors [23], [24] through years of continuous development and become synonymous with the real-time object detector. Though YOLO detectors achieve a reasonable tradeoff between accuracy and speed, they still rely on NMS postprocessing which is not trainable to reduce numerous overlapping boxes, making it unstable in both speed and accuracy due to NMS's hyperparameters.

The core reason CNN decoder-based detectors necessitate NMS is the one-to-many matching strategy employed in the label assignment phase. Recent Transformer decoder-based detectors do not suffer from the above issue as the one-to-one matching strategy for label assignment. DETR [7] first introduces a Transformer into the area of object detection, making it the first truly end-to-end detector. However, even with an NMS-free framework, the high computational cost prevents it from meeting real-time detection requirements. Due to its streamlined pipeline and elimination of hand-crafted components, DETR has received extensive attention from academia. Deformable DETR [11] introduce multiscale features and deformable operation into the encoder to accelerate DETR's training convergence. DN-DETR [8] and DINO [10] propose novel initial query selection schemes to obtain easy-to-optimize object queries in the decoder. To expeditiously process multiscale features, RT-DETR [13] designs an efficient hybrid encoder to replace the vanilla Transformer encoder, which significantly improves inference speed and extends DETRs to real-time scenarios. Despite its superior performance in natural image object detection, whether RT-DETR can be extended to the detection task within complex SAR imagery is less exploited.

B. Denoising in SAR Imagery

SAR imagery is often plagued by speckle noise, which can obscure crucial details and hinder object detection. Various denoising techniques have been developed to enhance the quality of SAR images, improving downstream tasks such as target recognition and classification.

Studies [25], [26] have consistently highlighted the detrimental effects of speckle noise on image interpretation and analysis, which is particularly pronounced in SAR images due to their unique acquisition process.

Traditional denoising methods, including spatial filtering and wavelet transforms, have been widely used in SAR imagery. For instance, techniques like the nonlocal means (NLMs) filter [27] leverage the redundancy in image data to reduce noise while preserving edges and textures. In recent years, deep learning-based methods have emerged as powerful tools for denoising SAR imagery. For example, MRD-DANet [28] employs a multiresolution approach to effectively mitigate noise while maintaining essential image details.

Both nondeep learning and deep learning-based denoising algorithms have been proposed and effectively applied to SAR imagery, showing significant improvements in downstream tasks by reducing noise-related ambiguities [28], [29].

However, the application of denoising strategies to SAR object detection has not been extensively explored, which is the focus of this article.

C. Deep Learning-Based SAR Object Detection

Various popular deep learning-based detection frameworks have demonstrated remarkable performance in natural image object detection. Nevertheless, object detection in SAR imagery presents distinct challenges, such as the small size of objects, the presence of speckle noise, and the background clutter. These unique characteristics make adopting generic object detectors off the shelf typically inadequate for SAR imagery. Therefore, contemporary deep learning-based SAR object detectors have predominantly concentrated on the development of network architectures or modules to tackle these specific issues. For instance, methods such as MGCAN [14], MSSDNet [30], and SEFEPNet [16] aim to bolster object feature representation through the integration of features across multiple scales. Fitnet [15] proposes a feature interaction module to facilitate comprehensive multiscale feature interaction, thereby mitigating the effects of noise and aligning features from various object scales. SRT-Net [31] introduces the scattering region topological structure pyramid to offer richer semantic information. Attention-based mechanisms are utilized in PADN [32] and EWFAN [33] to refine object features in the context of the speckle noise. CenterNet++ [34], building upon the foundation of CenterNet [35], further enhances robustness for SAR object detection by incorporating feature enhancement, multiscale fusion, and head refinement. In addition, CRTransSar [36] focuses on improving feature representation through advanced context representation learning.

Despite the above methods achieving significant progress, most of them concentrate on mitigating speckle noise through network architecture improvement, they ignore the noise issue at the level of input data. Furthermore, many studies pursue high accuracy with sacrificed real-time performance. Instead, we seek to address this dilemma through a carefully designed multilevel denoising strategy.

III. Proposed Method

To achieve a reasonable tradeoff between speed and accuracy for SAR object detection, a novel framework called the MD-DETR is constructed. Our framework is built upon the real-time end-to-end RT-DETR detector with three core denoising designs, that is, image-level denoising, feature-level denoising, and query-level denoising. Note that the proposed multilevel denoising strategy is general and can be integrated into other detectors, which is not the focus of this article.

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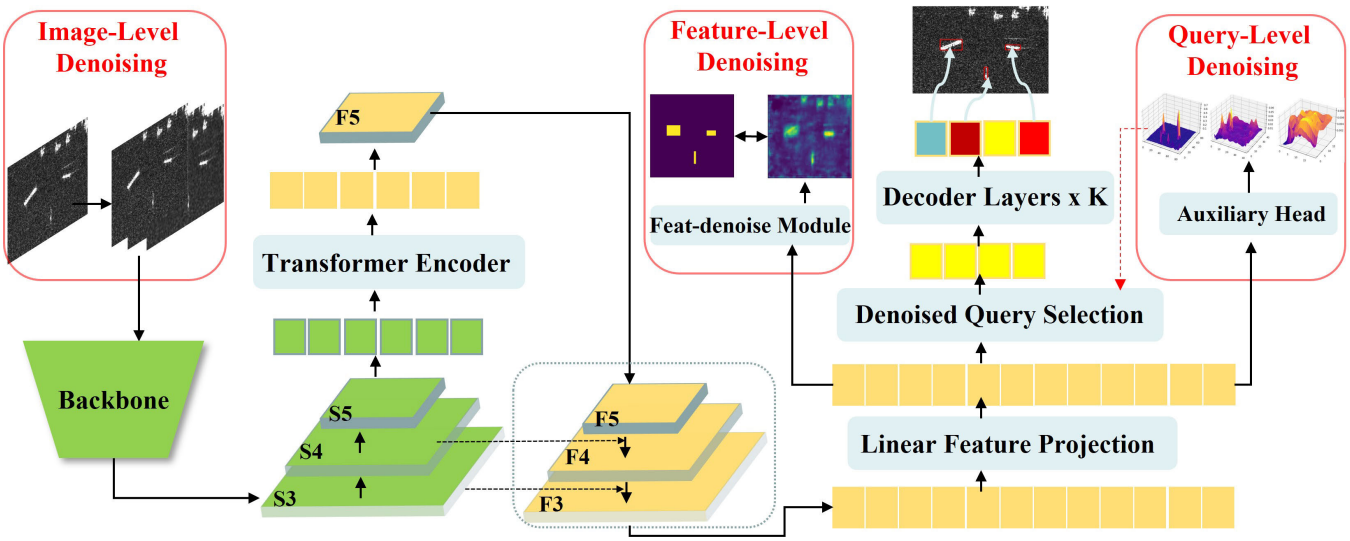


Fig. 1. Overall architecture of the proposed MD-DETR, which is built upon the RT-DETR detection framework with three core denoising designs, that is, the image-level denoising, the feature-level denoising, and the query-level denoising.

In this section, a detailed introduction to RT-DETR (the baseline in our experiments) and overall architecture is provided first. Next, we will dive into the technical details of the proposed three core denoising designs individually. Finally, the total loss is introduced in the training and inference part.

A. Overall Architecture

The real-time end-to-end RT-DETR detection framework is chosen as our preliminary network structure. Its overall architecture is shown in Fig. 1, which mainly consists of a backbone, a hybrid encoder, and a Transformer decoder.

1) Backbone: Given an image as input, a backbone with multistage (e.g., ResNet [37] and Swin Transformer [38]) may generate several feature maps with different resolutions from the last three stages of the backbone, and then the multiscale feature maps $\{S_3, S_4, S_5\}$ downsampled by 8, 16, and 32 with respect to the input image are fed into the encoder.

2) Hybrid Encoder: The efficient hybrid encoder consists of two modules, namely the attention-based intrascale feature interaction (AIFI) and the CNN-based cross-scale feature fusion (CCFF), where AIFI adopts a Transformer encoder to preform the single-scale interaction only on high-level S_5 feature-map with richer semantic concepts, and CCFF performs CCFF in the PAN [39] style. Instead of general convolution, the fusion block is composed of RepConv [40]. In summary, the hybrid encoder can be formulated as

$$\begin{aligned} Q = K = V &= \text{Flatten}(S_5) \\ F_5 &= \text{Reshape}\left(\text{softmax}\left(\frac{\text{QK}^T}{\sqrt{d_k}}\right)V\right) \\ M &= \text{CCFF}(\{S_3, S_4, F_5\}) \end{aligned} \tag{1}$$

where Reshape represents restoring the shape of the flattened feature to the same shape as S_5 and M represents the final encoded memory for the decoder.

3) Transformer Decoder: Based on the encoded feature M, a query selection schema is adopted to initialize content and positional queries for the decoder. With the initialized queries, deformable attention in cascaded decoder layers is used to combine the encoded feature M and the queries are updated layer-by-layer. The final outputs are formed with refined positional queries along with classification results predicted by refined content queries.

B. Image-Level Denoising

As discussed in Section I, most existing SAR detectors simply replicate the single-channel SAR image to the three-channel RGB-like image to accommodate pretrained models from ImageNet. However, this straightforward replication does not yield any additional information gain, making it a suboptimal approach. Thus, we propose an image-level denoising approach that leverages traditional image-denoising algorithms, such as Gaussian filtering and bilateral filtering, to perform denoising at the pixel level. Given the single-channel SAR image x, the generated denoised inputs D_i can be generally defined as

$$D_i = T_i(x), i \in \{\text{Gaussian filter, Bilateral filter}\} \tag{2}$$

where T_i is a predefined transformation. In our study, we choose two filtering methods for denoising due to their distinct, yet complementary advantages in addressing the speckle noise prevalent in SAR imagery.

We opted for Gaussian and bilateral filtering due to their distinct, yet complementary advantages in addressing the speckle noise. Gaussian filtering applies a weighted average to each pixel based on a Gaussian distribution, which is effective in reducing noise while preserving the overall structure of the image. However, it may sometimes result in the blurring of edges and fine details. To counteract this potential drawback, we employed bilateral filtering, which is an edge-preserving smoothing technique. It reduces noise while maintaining edges and important details by applying a weighted average that considers both the spatial proximity and intensity similarity of pixels.

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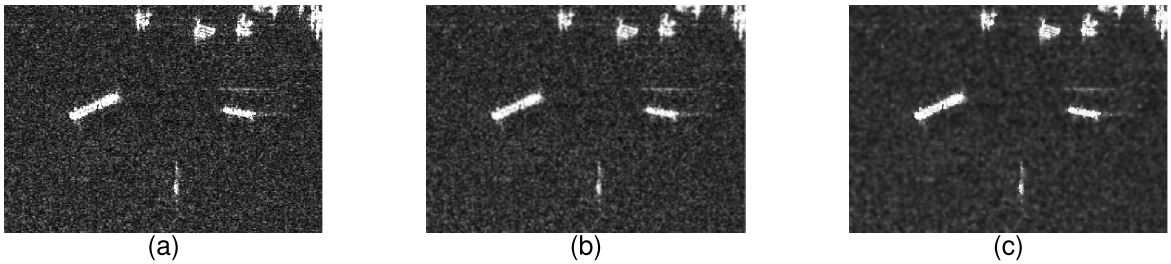


Fig. 2. Example of single-channel SAR image and its corresponding denoised counterparts. (a) Original SAR image. (b) Gaussian filtering one. (c) Bilateral filtering one.

By combining these two filters, we aimed to leverage their respective strengths to achieve a balanced denoising effect. This dual-filter approach ensures that the denoised image retains the necessary structural information required for accurate object detection in SAR images.

Consequently, the two denoised images are concatenated with the original image along the channel dimension, forming a genuine three-channel image X that serves as input to the detection model. It can be defined as

$$X = \text{concat}(x, D_i). \tag{3}$$

An example of a single-channel SAR image and its corresponding denoised counterparts is presented in Fig. 2. As can be seen from Fig. 2, the Gaussian filter provides an initial reduction of noise, while the bilateral filter refines the process by preserving edges and details. This dual-filter approach ensures that the denoised image retains the necessary structural information for accurate object detection.

C. Feature-Level Denoising

Different from natural images, SAR imagery exhibits the speckle noise characteristic, which inevitably results in encoder features with low discriminative and localization confidence, and inspires us to perform denoising at the feature level.

Specifically, as depicted in Fig. 1, we propose a lightweight feature denoising module (FDM) appended on the encoded feature after linear feature projection. The feature from F_3 is chosen as the input to FDM as the lower-level feature with higher-resolution is beneficial for small objects common in SAR imagery. The FDM simply consists of a 1×1 convolution layer with batch normalization, the flattened features are first reshaped to restore the shape as F_3 and then processed by FDM, and the output score map indicates whether there is an object at each pixel. In the training phase, the annotated bounding boxes are used to generate the 0–1 mask, where pixels encompassed within the bounding boxes are labeled 1 and the others are assigned a label as 0. We use this generated pseudo-mask as the ground truth to supervise the predicted score map. Detail illustration is depicted in Fig. 3.

As there is a multitude of background pixels in this task, to maximize the scores where there is an object, we employ focal loss [24] for the gradient-based optimization of FDM

$$L_{\text{fld}} = \sum_i -\alpha_i (1 - p_i)^\gamma \log(p_i), i \in \{0, \ldots, HW\} \tag{4}$$

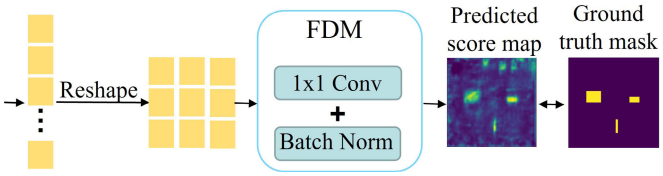


Fig. 3. Illustration of the proposed feature-level denoising pipeline. A lightweight FDM is adapted to translate the input feature map into the predicted score map.

where $p_i \in [0, 1]$ is the predicted probability indicating whether there is an object in each location i, H and W are the height and width of the ground-truth mask, respectively, and α and γ are the focusing hyperparameters to control the penalty of different samples.

By incorporating this coarse-mask guidance learning task, the detector is directed to focus more on areas corresponding to object locations while simultaneously suppressing background noise. It is worth noting that the FDM is solely utilized during training, where it plays a pivotal role in facilitating the learning process by enhancing the model's ability to distinguish between objects and background noises. Once the training is complete, it is entirely discarded during inference. Consequently, it does not incur any additional computational overhead, making it an economically efficient and effective solution for performance gain.

D. Query-Level Denoising

To reduce the difficulty of optimizing initial queries in DETR, RT-DETR proposes an uncertainty-minimal query selection scheme, which uses both the confidence and location scores to select the top K positional and content queries from the encoder feature as initial queries. Despite its high quality, RT-DETR necessitates a fixed number (e.g., 300) of queries per image to achieve higher recall. Nevertheless, the number of objects in each SAR image can vary significantly, and the fixed quota of initial queries leads to unnecessary computational overhead. In addition, the query selection in RT-DETR relies on a one-to-one matching strategy during training, resulting in suboptimal initialization for the decoder.

To address this problem, we propose a query-level denoising strategy on dynamically selecting a smaller, yet more accurately localized initial query. Specifically, an auxiliary head with the one-to-many matching strategy is proposed to support flexible query selection, in which the number of queries is

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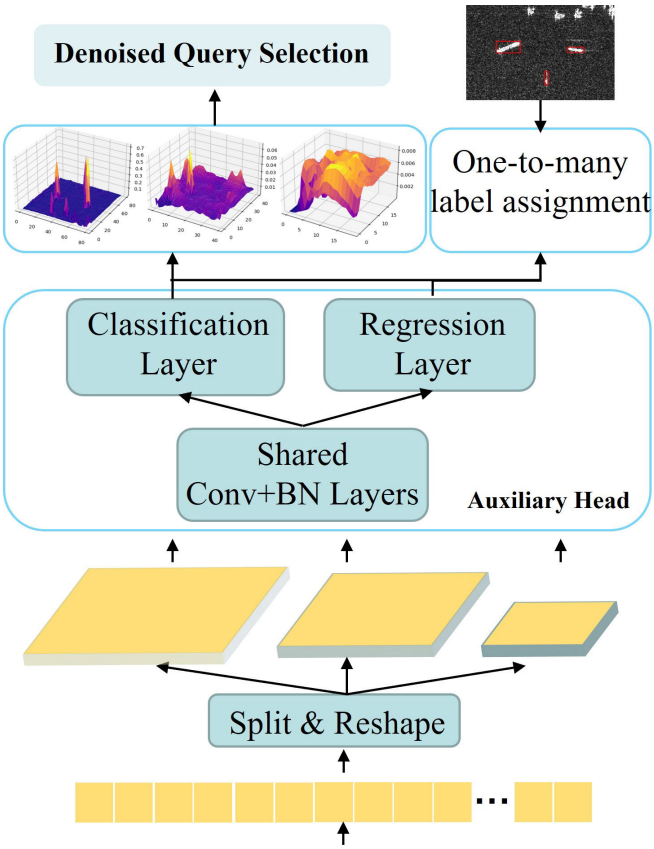


Fig. 4. Illustration of the proposed query-level denoising pipeline. The encoder feature after linear projection is first split and reshaped to the same shape as $\{F_3, F_4, F_5\}$. A lightweight auxiliary head with shared convolution layers encodes these features for classification and regression. Finally, the output of the classification layer is used to generate the dynamic number of initial queries. During training, the outputs of both classification and regression are used for one-to-many label assignment.

dynamically adjusted based on the object density within an image. Technical illustration is presented in Fig. 4. As can be seen from Fig. 4, the encoder feature after linear projection is first split and reshaped to the same shape as $\{F_3, F_4, F_5\}$ and then a lightweight auxiliary head is built to generate the object density within an image. For simplicity, we use four shared convolution layers with batch normalization to encode the features for classification and regression. Both the classification layer and the regression layer consist of a convolution layer. The output of the classification layer can be exhibited as the object density maps, which have the same shape as the three input feature maps. Based on the object density maps, we calculate the dynamic number of initial queries for each image in a simple way

$$\text{num}_q = \min\left(\max\left(q_{\min}, \sum_i \sum_{x,y} d(x,y), q_{\max}\right)\right) \tag{5}$$

where i indicates the index of the object density map, there are totally three density maps in outsetting, and $d(x, y)$ means the density value is each location of the density maps. $q_{\min}$ and $q_{\max}$ are the minimum and maximum number of queries in each image, respectively. Experimentally, we set $q_{\min}$ and $q_{\max}$ as 30 and 300, respectively. More details can be seen in our experiments (Section IV-C).

For training, we adopt the one-to-many dynamic label assignment as the same as that in TOOD [41], where each ground-truth box can match with multiple anchors according to the dynamic IoU threshold. It is beneficial for selecting positive samples from different feature maps automatically. For bounding box regression, we adopt the relative offsets from the location to the four sides of a bounding box as the regression target. For the joint learning of classification and regression tasks, we choose varifocal loss (VFL) [42] and distribution focal loss (DFL) [43], respectively. Different from the quality focal loss (QFL) in [43], VFL uses target score to weight the loss of positive samples, making positive samples with high IoU contribute relatively largely to the loss, thus guiding the model to pay more attention to high-quality samples. DFL proposes to use general distribution to predict the bounding box, thus being more flexible in bounding box representation. In addition, we also append the GIoU loss for better localization. Finally, our lightweight auxiliary head is supervised by the following loss function:

$$L_{\text{qld}} = \frac{1}{N_{\text{pos}}}(\alpha \cdot \text{loss}_{\text{VFL}} + \beta \cdot \text{loss}_{\text{DFL}} + \gamma \cdot \text{loss}_{\text{GIoU}}) \tag{6}$$

where N_{pos} is the number of positive samples, and α, β, and γ are the weights for VFL, DFL, and GIoU losses, respectively.

E. Training and Inference

The above three denoising designs constitute the proposed MD-DETR. During the training phase, we employ a multitask learning procedure. Specifically, the backbone, the hybrid encoder, and the Transformer decoder are optimized jointly along with the proposed FDM and the auxiliary head in an end-to-end fashion. The overall loss function is comprised of three components, that is, the Transformer decoder loss L_{decoder} which is the same as that in RT-DETR, the feature-level denoising loss (in 4) and the query-level denoising loss (in 6)

$$\text{Loss} = L_{\text{decoder}} + L_{\text{fld}} + L_{\text{qld}}. \tag{7}$$

During the inference phase, the FDM is absolutely abandoned, but the lightweight auxiliary head is reserved to generate a dynamic number num_q (in 5) of initial queries for the query selection step. In this way, each image will have a different number of initial queries for the decoder to refine. Though the average number of initial queries is particularly much less than the fixed quota, the detection accuracy is maintained. Quantitative results can be seen in our experiments (see Section IV-C).

IV. Experimental Results and Analysis

In this section, we present the experimental results and analysis. The details of the benchmark datasets are presented in Section IV-A. Ablation experiments in Section IV-C are conducted on the SSDD dataset, and comparisons with the state-of-the-arts (SOTAs) on both SSDD and SARDet-100K are reported in Section IV-D. For all experiments, we use the standard COCO AP metric for evaluation with single-scale inference.

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TABLE I
IMAGE AND INSTANCE LEVEL STATISTICS OF THE SSDD AND SARDET-100K DATASETS

Dataset	Images		Instances		Ins/Img
	Train	Val	Train	Val	
SSDD [17]	928	232	2041	546	2.23
SARDet-100K [1]	94,493	10,492	198,747	22,703	2.11

A. Dataset

To evaluate the effectiveness of the proposed MD-DETR, we conduct a series of experiments on two benchmark datasets, that is, the SSDD [17] and the recently published COCO-level large-scale multiclass SAR object detection dataset (SARDet-100K) [1]. Below is the detailed information about the dataset.

SSDD [17]: SSDD is the first open dataset that is widely used to research SOTA detectors for SAR ship detection based on deep learning. It contains multiscale ship slices from RADARSAT-2, TerraSAR-X, and Sentinel-1 satellites, with a total of 1160 images and 2456 annotated ship objects. The image resolution varies from 1 to 15 m. We adopt the bounding box annotations of the SSDD's official release version, and the official training-test division for model training and evaluation, with 928 and 232 images, respectively. The SSDD dataset further divides the test images into offshore and inshore types, with 186 and 46 images, respectively. In this article, the detector is trained on the training set and tested under these two scene types to verify the detection performance under simple samples (offshore scene) and difficult samples (inshore scene).

SARDet-100K [1]: SARDet-100K is a new benchmark dataset for large-scale SAR object detection, which is the first COCO-level large-scale multiclass SAR object detection dataset. It encompasses a total of 116 598 images and 245 653 instances. Compared to SSDD only for ship detection, SARDet-100K covers six categories, that is, ship, aircraft, car, bridge, harbor, and tank. The scale-level and category-level diversity of the SARDet-100K dataset provides detectors with robust training and evaluation. These images are collected from different countries and institutions, making it a much more diverse dataset for benchmarking. It contains 94 494 images for training and 10 492 images for validation. As the annotations for the test set in not available, we use the training set for training and the validation set for evaluation.

Detailed information on the two benchmark datasets is shown in Table I.

B. Implementation Details

We implement the proposed method in PyTorch based on the RT-DETR official repository.[1] As for the tradeoff of speed and accuracy, the backbone is chosen as ResNet-50 [37]. Adam is applied to optimize the network. We also apply the strategy of moving average weights as in [13] to achieve more stable training. The cosine learning rate schedule is adopted and the base learning rate is 0.0001. The batch size is 8 per Nvidia RTX 4080 GPU device. For SSDD, models are trained for 12 epochs on a single GPU. For SARDet-100K, models are trained for 36 epochs on two Nvidia RTX 4080 GPUs as the dataset is much larger, a longer training duration is required for model convergence. Specifically, SSDD training duration is approximately 12 min, while SARDet-100K requires about 36.5 h. For both datasets, models are initialized from COCO pretrained. We only use some basic data augmentations, including random crop, random horizontal flip, and random color distortion. The input size is resized to fixed 640×640 for both training and testing. We apply single-scale evaluation on both datasets for speed consideration.

[1]https://github.com/lyuwenyu/RT-DETR

C. Ablation Study

We conduct a series of ablation studies on the SSDD dataset to evaluate the individual contributions of each denoising design and the overall performance of the proposed MD-DETR. For a fair comparison, consistent settings are maintained for all experimental settings, each experiment has five runs of which the average is reported in Table II. In Table II, the ID 0 means the original RT-DETR network, which serves as our baseline. Subsequently, the proposed multilevel denoising strategies are gradually added to the baseline, represented by ID 1, 2, and 3, respectively. Compared to the baseline, each of our proposed multilevel denoising strategies significantly contributes to the performance gain, with an improvement of 0.4, 0.5, and 0.2 in AP gradually. All three denoising strategies constitute the proposed MD-DETR, achieving the best performance in both AP and AP_{50}. Utilizing the same GPU device of Nvidia RTX 4080, RT-DETR runs at 44.12 ms per image, the image-level denoising requires no computation burden with a performance gain of 0.4, and the feature-level denoising has an additional performance gain of 0.4 while running at 44.84 ms per image. The final MD-DETR outperforms the baseline by 1.1% in AP and 0.8% in AP_{50}, while with a minor additional number of parameters and still runs at a speed of 46.06 ms per image.

It is worth noting that, image-level denoising hurts the detection performance for small objects. However, equipped with the feature-level and query-level denoising, the final MD-DETR also achieves a comparable performance to the baseline, with 75.6 in AP_{small}.

As is known to all, the inshore ships with complex backgrounds are more challenging to detectors, thus we also evaluate the proposed MD-DETR under these two scenes to verify the performance under simple samples (offshore scene) and difficult samples (inshore scene). Evaluation results are given in Table III. Table III shows that the proposed MD-DETR beats the baseline in both scenes, which also indicates that detection in the inshore scene still has considerable room for improvement compared to that in the offshore scene, which indicates that the proposed method can better handle background clutter and are more robust in complex backgrounds.

These results in Tables II and III clearly demonstrate that the proposed multilevel denoising strategies enhance the

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TABLE II
ABLATION STUDIES OF THE PROPOSED MULTILEVEL DENOISING

ID	Multi-level denoising			AP	AP_{50}	AP_{75}	AP_{small}	#Params(M)	Test time(ms/img)
	Image	Feature	Query						
0	×	×	×	74.6	98.1	93.4	75.6	42.73	44.12
1	✓	×	×	75.0(+0.4)	98.2	93.2	74.8	42.73	44.12
2	✓	✓	×	75.5(+0.5)	98.5	92.4	75.1	42.73	44.84
3	✓	✓	✓	**75.7**(+0.2)	**98.9**	93.2	75.6	43.91	46.06

TABLE III
COMPARISON WITH THE BASELINE IN THE INSHORE AND OFFSHORE SETS

Method	AP		AP_{50}	
	Inshore	Offshore	Inshore	Offshore
Baseline	73.3	75.8	95.7	99.1
MD-DETR(ours)	74.3	77.0	96.5	99.8

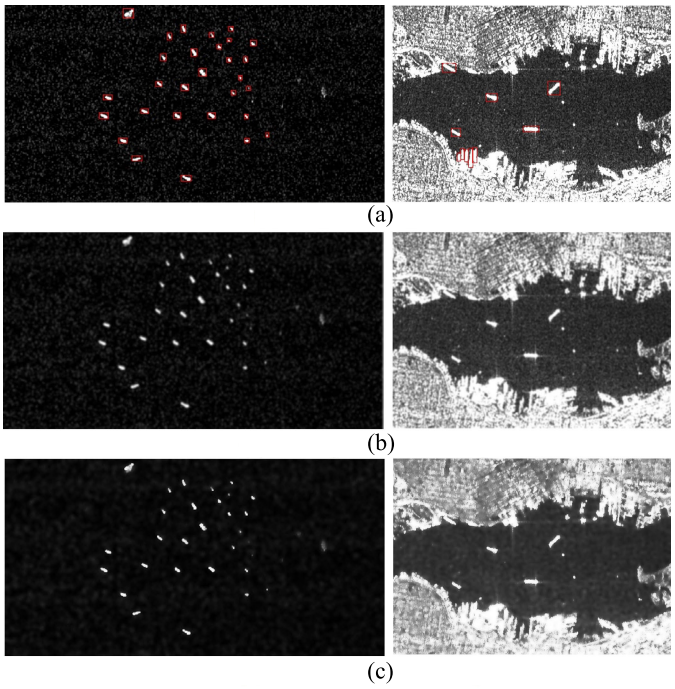


Fig. 5. Two examples of the proposed image-level denoising. (a) Original SAR image, in which the ship objects are encompassed by the red bounding box. (b) Image after Gaussian filtering. (c) Image after bilateral filtering.

detection accuracy of the baseline. Next, we will analyze the contribution of each module in detail.

1) Effect of image-level denoising: The proposed image-level denoising approach leverages traditional image-denoising algorithms, such as Gaussian filtering and bilateral filtering to generate denoised images. Apart from Fig. 2, more examples are given in Fig. 5, where two original images are from the offshore and inshore scene, respectively. We use the red bounding box to highlight the ship objects in the original images. As can be seen from Figs. 2 and 5, Gaussian filtering

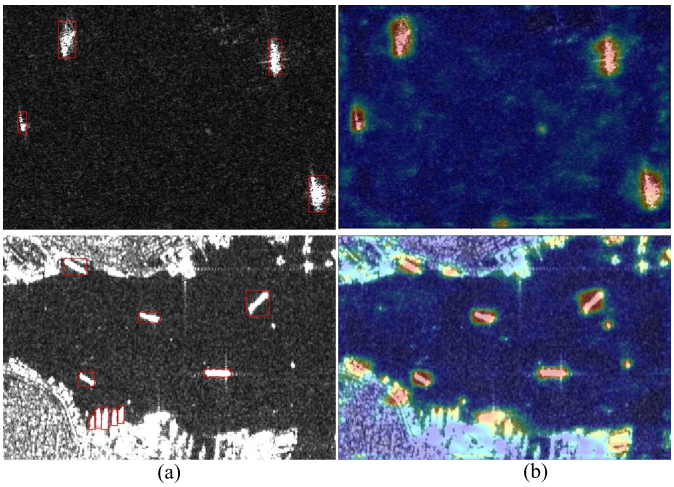


Fig. 6. Two examples of the proposed feature-level denoising. (a) Original SAR image, in which the ship objects are encompassed by the red bounding box. (b) Score map predicted by the proposed FDM in Section III-C.

is helpful to reduce speckle noise which may blur the edges. Bilateral filtering is capable of maintaining the edges and import details. By combining these two filters, we leverage the strengths of each to achieve a balanced denoising effect. Subsequently, the denoised image is concatenated with the original image along the channel dimension, forming a genuine three-channel image for training and inference. We also tried other image-denoising algorithms, like mean filtering and median filtering, and experimental results indicate that the differences are minimal, suggesting that more advanced image-level denoising techniques required further exploration, which is not the focus of this article.

2) Effect of feature-level denoising: Table II shows that the feature-level denoising can further improve the AP by 0.5, compared to the baseline with denoised image input. Fig. 6 gives more evidence to demonstrate its effectiveness, where two original images are from the offshore and inshore scenes, respectively. We use the red bounding box to highlight the ship objects in the original images, and the predicted score map from the FDM in Section III-C is overlayed on the original image. As illustrated in Fig. 6, the trained model pays more attention to the locations where the objects lay when the FDM is applied. In this way, the foreground objects become more salient while the background noise is significantly suppressed during training, leading to a more discriminative and robust

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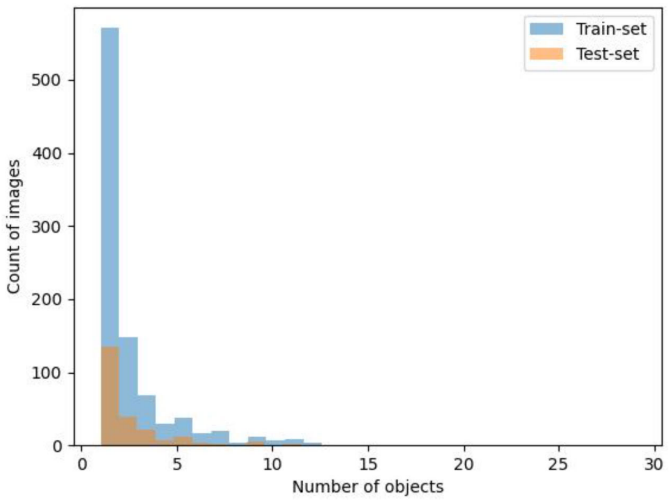


Fig. 7. Distribution of the number of objects in each image of the SSDD dataset.

TABLE IV
COMPARISONS OF DIFFERENT QUERY SELECTION STRATEGIES

Query Selection	#Queries	AP	AP_{50}
Fixed number	30	74.1	97.2
	300	75.5	98.5
Dynamic number	-	75.7	98.9

feature representation for the transformer decoder. Note that the FDM is discarded during inference, thus incurring no additional computational overhead, making it economically efficient and effective for performance gain.

3) Effect of query-level denoising: Most Transformer decoder-based detectors necessitate a fixed number (K) of queries per image in the query selection part. For example, $K = 300$ is the default setting in RT-DETR. However, the quantity of objects in each SAR image can vary significantly, and the static allocation of a large number of queries results in excessive computation for images with fewer objects. Taking the SSDD dataset as an example, the number of objects in each SAR image is relatively small as depicted in Fig. 7, most images have only one object, the mean and median number of objects in both the training set and the test set is 2 and 1, the maximum number in the training set and the test set is 29 and 24, respectively. It indicates that a large number of fixed queries is suboptimal. We also tried a small number of queries (with $K = 30$). From the first row in Table IV, it is found that directly decreasing the number of queries can significantly hurt the performance, the AP with fixed 30 initial queries decreases to 75.5 from 74.1, compared to the baseline with 300 queries. While equipped with the proposed query-level denoising, the number of queries per image is dynamically adjusted according to the object density map, we achieved a performance gain of 0.2 in AP even with fewer queries.

To better visualize the variation of the query number during training, Fig. 8 presents the average query number per batch through the total training process. With the query-level denoising, the query number is varied (see 5) during training. On the

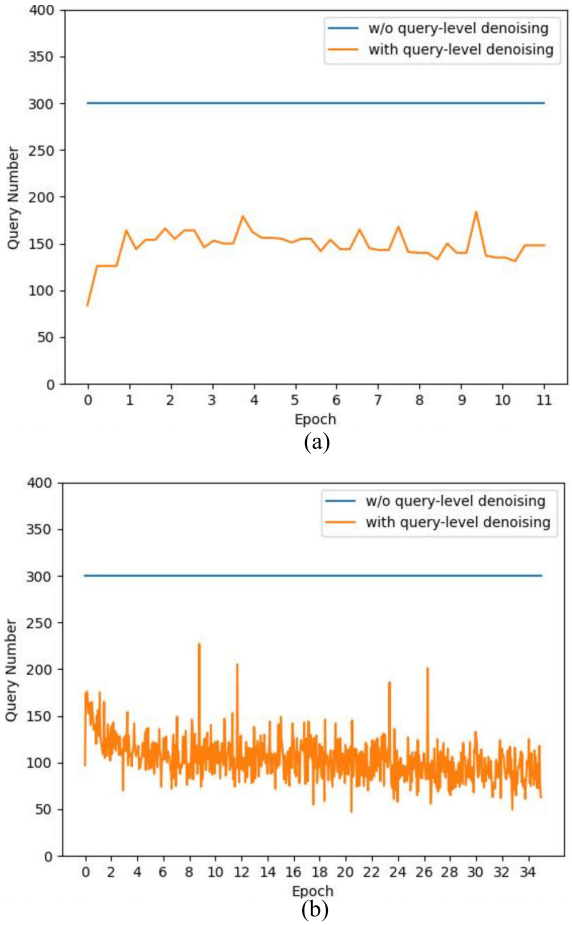


Fig. 8. Variation of average query number per batch through the total training process, in (a) SSDD and (b) SARDet-100k, respectively.

contrary, the baseline model has a fixed number of 300 strictly. Fig. 8(a) shows this process in the SSDD dataset with a total training epoch of 12. In the first epoch, the average query number is less than 100, which is because the auxiliary head for query-level denoising is not well optimized. As the training progresses, the query number remains stable at around 150, which is only about half of the fixed quota in the baseline. Furthermore, we also plot the variation during training in SARDet-100K in Fig. 8(b) as it exhibits a longer training duration of 36 epochs. It is interesting to see the query number steadily decreasing as the training progresses, which indicates that the auxiliary head is more confident about the foreground object locations. Finally, the query number in SARDet-100k remains around 100, much less than the fixed quota of 300. Therefore, we set the maximum number of queries in (5) as 300. To avoid the extreme case with no objects, we set the minimum number of queries in (5) as 30.

4) Overall effect of multilevel denoising: To further demonstrate the overall effect of the proposed multilevel denoising, we depicted several final detection results on the test set, compared to the baseline. As shown in Fig. 9, the baseline method cannot handle false alarms well due to the speckle noise in SAR images. For example, in Fig. 9(a), large ship objects with extreme aspect ratios are challenging for accurate detection. Equipped with the proposed query-level denoising,

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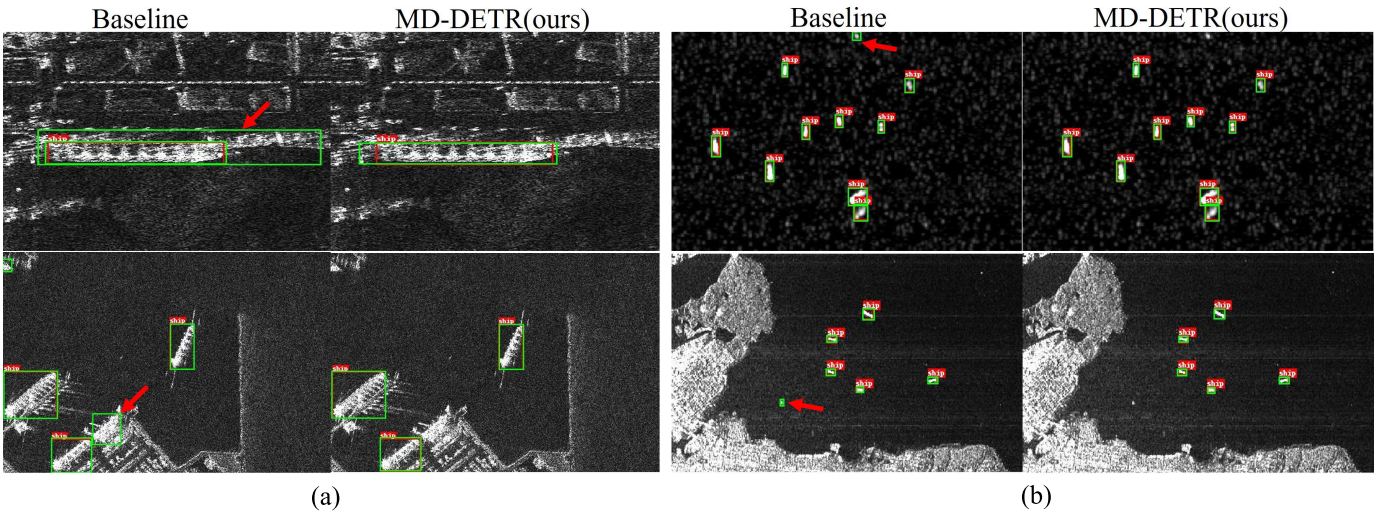


Fig. 9. Comparison results of the proposed MD-DETR and the baseline in the SSDD dataset. Red and green bounding boxes represent the ground truth and the predictions from the detector, respectively. Red arrows indicates the false alarms. (a) False alarms of large objects. (b) False alarms of small objects.

our MD-DETR can give a high-quality bounding box without duplicate detection. In the second row of Fig. 9(a), MD-DETR is also able to reduce false alarms with a similar appearance to true positives. For small objects in Fig. 9(b), the baseline method is vulnerable to the background speckle noise. However, thanks to the overall effect of multilevel denoising, MD-DETR is capable of detecting small objects, maintaining robust performance in strong speckle noise.

D. Comparison With SOTAs

To better show the superiority of the proposed MD-DETR, we compare various SOTA methods on both the SSDD and SARDet-100k datasets.

Table V presents a comprehensive comparison of the proposed MD-DETR with several SOTA methods on the SSDD dataset. The results are color-coded, with red highlighting the top performance and green for the second best. For general detectors, we report the results of the first seven methods from the RSPrompter [63] paper and the last seven methods from the MSFA [1] paper. For SAR detectors, we report the results of the first ten methods from the MSFA [1] paper and the last three from the RSPrompter [63] paper. From Table V, it is evident that MD-DETR outperforms all other methods for both general detection and SAR detection across all three metrics. With an AP of 75.7, AP_{50} of 98.9, and AP_{75} of 93.2, MD-DETR demonstrates its superiority in detecting objects with higher precision. It surpasses the second-best performer, MSFA [1] built upon faster R-CNN, by 1.0 in AP_{50}. It was noted that MSFA is tested with the input image size of 800×800, while the proposed MD-DETR is with a lower image size of 640×640, running at 46.06 ms per image at the Nvidia RTX 4080 GPU device.

Table VI presents a detailed comparison of the proposed MD-DETR with various SOTA methods on the SARDet-100k dataset. The metrics for evaluation include mean average precision (mAP), mAP at an IoU threshold of 0.5 (mAP_{50}), mAP at 0.75 (mAP_{75}), and mAP for small, medium, and large objects (mAP_{small}, mAP_{medium}, and mAP_{large}), which provide a comprehensive understanding of the detectors' performance across different object sizes. We report the results of all competitors from the MSFA [1] paper, where all the two-stage,

TABLE V

COMPARISONS OF THE PROPOSED MD-DETR WITH PREVIOUS SOTA METHODS ON SSDD. RED AND GREEN INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Detectors		AP	AP_{50}	AP_{75}
General Detectors	Mask R-CNN [44]	67.7	95.6	84.9
	MS R-CNN [45]	67.8	95.3	89.9
	HTC [4]	69.3	97.1	85.7
	SCNet [46]	66.9	92.5	82.5
	CondInst [47]	68.1	92.4	85.5
	BoxInst [48]	62.8	96.2	74.7
	Mask2Former [49]	62.7	90.7	75.6
	Grid R-CNN [50]	-	88.9	-
	Faster R-CNN [2]	-	89.7	-
	Cascade R-CNN [3]	-	90.5	-
	Free-Anchor [51]	-	91.0	-
	Double-Head R-CNN [52]	-	91.1	-
	PAN [39]	-	91.2	-
	DCN [53]	-	92.3	-
SAR Detectors	NNAM [54]	-	79.8	-
	DCMSNM [55]	-	89.6	-
	ARPN [56]	-	89.9	-
	DAPN [57]	-	90.6	-
	HR-SDNet [58]	-	90.8	-
	SER Faster R-CNN [59]	-	91.5	-
	FBR-Net [60]	-	94.1	-
	CenterNet++ [34]	-	95.1	-
	CRTransSar [36]	-	97.0	-
	Faster R-CNN + MSFA [1]	-	97.9	-
	CATNet [61]	67.5	96.8	80.4
	HQ-ISNet [62]	66.6	95.9	80.2
	RSPrompter [63]	70.4	97.7	86.2
	MD-DETR(ours)	75.7	98.9	93.2

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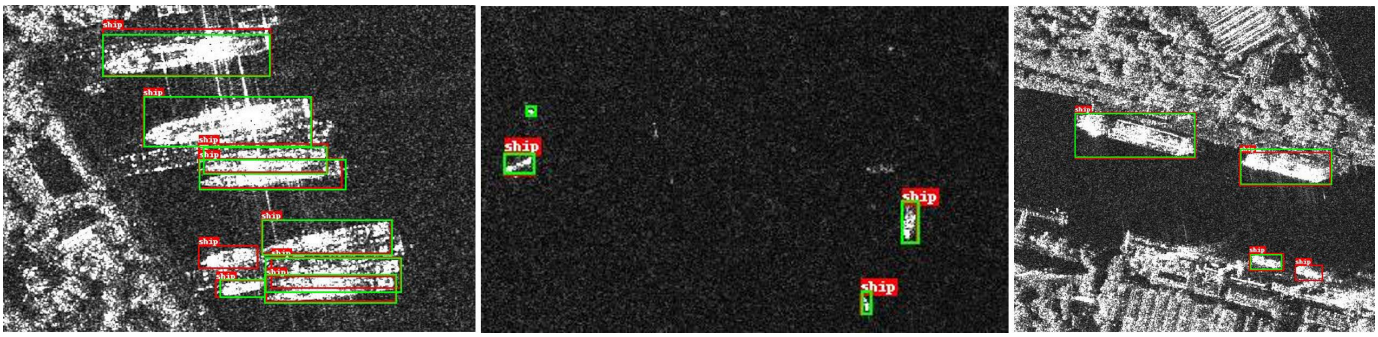


Fig. 10. Failure cases of SAR ship detection in the SSDD dataset. Red and green bounding boxes represent the ground truth and the prediction from the proposed MD-DETR, respectively.

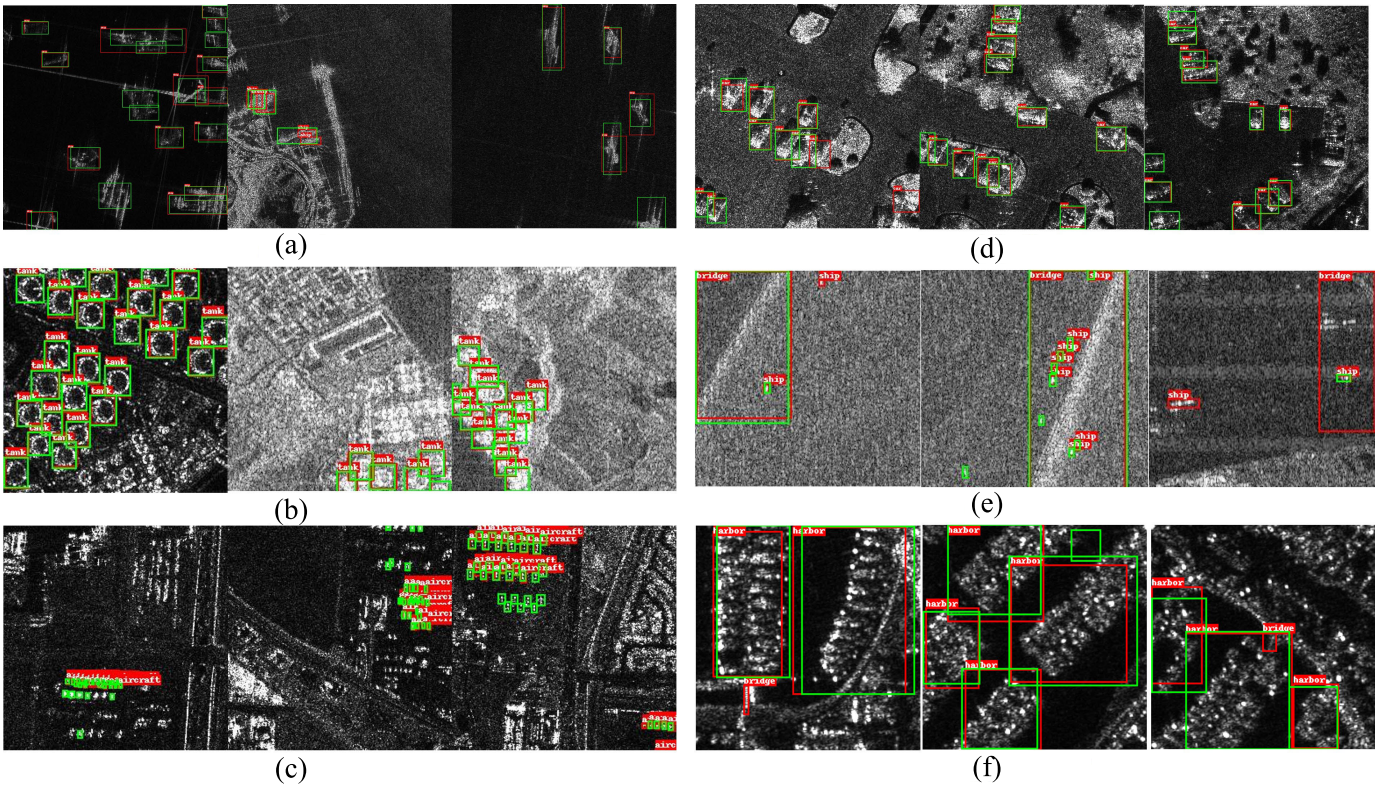


Fig. 11. Failure cases of multiclass object detection in the SARDet-100k dataset. Red and green bounding boxes represent the ground truth and the prediction from the proposed MD-DETR, respectively. (a) Ship. (b) Tank. (c) Aircraft. (d) Car. (e) Bridge. (f) Harbor.

single-stage, and end-to-end approaches are trained with the MSFA pretrain pipeline and are tested with the input size of 800 × 800. With an even smaller input size of 640 × 640, MD-DETR achieves the best across all metrics, signifying its superiority in multiclass SAR object detection. With an mAP of 57.2, mAP_{50} of 88.5, mAP_{75} of 62.8, and competitive performance on small (50.3), medium (68.3), and large (63.8) objects, MD-DETR demonstrates its effectiveness in handling a wide range of object sizes. The results from Table VI indicate that the proposed MD-DETR sets a new benchmark for object detection in the SARDet-100k dataset.

E. Limitations and Further Work

Our proposed method, MD-DETR, has achieved a tradeoff between high speed and accuracy in the task of SAR object detection. However, there are still some limitations that we acknowledge and discuss in this section.

For SAR ship detection, as illustrated in Fig. 10, our model occasionally struggles with detecting ships that are densely packed in the inshore scene or have similar signatures to the surrounding sea clutter. It is the same as in Fig. 11(a). This suggests that more techniques are required to tackle the complex background clutter.

For multiclass object detection in the SARDet-100k dataset, we plot the failure cases of all six categories. First, the categories of ship and aircraft (in Fig. 11(a), (c), and (e)) have more small objects. This suggests that more techniques are required to tackle the issue of small objects. Second, the failure cases in Fig. 11(b) and (d) indicate that blurred background may lead to a higher likelihood of missed positives and false alarms. Finally, as depicted in Fig. 11(e) and (f),

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TABLE VI
COMPARISONS OF THE PROPOSED MD-DETR WITH PREVIOUS SOTA METHODS ON SARDET-100K. RED AND GREEN INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Detectors		mAP	mAP_{50}	mAP_{75}	mAP_{small}	mAP_{medium}	mAP_{large}
Two Stage	Faster R-CNN + MSFA [1], [2]	51.1	83.9	54.7	45.2	62.3	57.5
	Cascade R-CNN + MSFA [1], [3]	53.9	83.4	59.8	47.2	66.1	63.2
	Grid R-CNN + MSFA [1], [50]	51.5	81.7	56.3	45.1	64.1	60.0
Single stage	RetinaNet + MSFA [1], [24]	49.0	80.1	52.6	41.3	61.1	59.4
	GFL + MSFA [1], [43]	53.7	84.2	57.8	47.8	66.2	59.5
	FCOS + MSFA [1], [64]	48.5	82.1	51.4	42.9	60.4	56.0
End to End	DETR + MSFA [1], [7]	47.2	77.5	49.8	37.9	62.9	58.2
	Deformable DETR + MSFA [1], [11]	51.3	85.3	54.0	44.9	65.6	61.7
	Sparse RCNN + MSFA [1], [65]	41.4	74.1	41.8	33.6	53.9	53.4
	Dab-DETR + MSFA [1], [66]	48.2	81.1	51.0	41.2	63.1	55.4
	MD-DETR(ours)	57.2	88.5	62.8	50.3	68.3	63.8

bridges and harbors pose a significant challenge due to their intricate structures with varying shapes and sizes, detectors may be confused by the complex radar reflections, which suggests that rotated object detection could be valuable for further development.

V. CONCLUSION

This article presents the MD-DETR, a novel framework for high-quality object detection in SAR images within complex scenes. By integrating three core denoising strategies: image-level, feature-level, and query-level denoising, we have demonstrated that MD-DETR can effectively mitigate the adverse effects of speckle noise inherent in SAR imagery. Our experimental results on two benchmark datasets, the SSDD and SARDet-100K, have shown that MD-DETR outperforms existing SOTA detectors, setting a new benchmark for SAR object detection. Despite the promising results, we acknowledge some limitations and challenges that still remain for further work, particularly in the context of handling small objects and improving the robustness of the model against background clutter. We believe that the MD-DETR represents a significant step forward in the field of SAR object detection and provides a foundation for future research.

REFERENCES

[1] Y. Li et al., "SARDet-100K: Towards open-source benchmark and ToolKit for large-scale SAR object detection," 2024, *arXiv:2403.06534*.

[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[3] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[4] K. Chen et al., "Hybrid task cascade for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4974–4983.

[5] J. Glenn. (2023). *YOLOv8*. [Online]. Available: https://github.com/ultralytics/ultralytics/tree/main

[6] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.

[7] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Glasgow, U.K.: Springer, 2020, pp. 213–229.

[8] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 13619–13627.

[9] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3651–3660.

[10] H. Zhang et al., "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[11] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[12] Y. Wang, X. Zhang, T. Yang, and J. Sun, "Anchor DETR: Query design for transformer-based detector," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 3, pp. 2567–2575.

[13] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," 2023, *arXiv:2304.08069*.

[14] L. Chen, R. Luo, J. Xing, Z. Li, Z. Yuan, and X. Cai, "Geospatial transformer is what you need for aircraft detection in SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5225715.

[15] Q. Hu, S. Hu, S. Liu, S. Xu, and Y.-D. Zhang, "FINet: A feature interaction network for SAR ship object-level and pixel-level detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5239215.

[16] P. Zhang et al., "SEFEPNet: Scale expansion and feature enhancement pyramid network for SAR aircraft detection with small sample dataset," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3365–3375, 2022.

[17] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[18] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[19] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[20] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[21] J. Glenn. (2022). *YOLOv5 Release V7.0*. [Online]. Available: https://github.com/ultralytics/yolov5/tree/v7.0

[22] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 7464–7475.

[23] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.* Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 21–37.

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[24] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[25] S. R. Cloude and E. Pottier, "A review of target decomposition theorems in radar polarimetry," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 2, pp. 498–518, Mar. 1996.

[26] J. Bruniquel and A. Lopes, "Multi-variate optimal speckle reduction in SAR imagery," *Int. J. Remote Sens.*, vol. 18, no. 3, pp. 603–627, Feb. 1997.

[27] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2005, pp. 60–65.

[28] S. Liu, Y. Lei, L. Zhang, B. Li, W. Hu, and Y.-D. Zhang, "MRDDANet: A multiscale residual dense dual attention network for SAR image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[29] S. Parrilli, M. Poderico, C. V. Angelino, and L. Verdoliva, "A nonlocal SAR image denoising algorithm based on LLMMSE wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 606–616, Feb. 2011.

[30] K. Zhou, M. Zhang, H. Wang, and J. Tan, "Ship detection in SAR images based on multi-scale feature extraction and adaptive feature fusion," *Remote Sens.*, vol. 14, no. 3, p. 755, Feb. 2022.

[31] D. Pan et al., "SRT-net: Scattering region topology network for oriented ship detection in large-scale SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5202318.

[32] Y. Zhao, L. Zhao, C. Li, and G. Kuang, "Pyramid attention dilated network for aircraft detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 662–666, Apr. 2021.

[33] J. Wang et al., "Integrating weighted feature fusion and the spatial attention module with convolutional neural networks for automatic aircraft detection from SAR images," *Remote Sens.*, vol. 13, no. 5, p. 910, Feb. 2021.

[34] H. Guo, X. Yang, N. Wang, and X. Gao, "A CenterNet++ model for ship detection in SAR images," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107787.

[35] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[36] R. Xia et al., "CRTransSar: A visual transformer based on contextual joint representation learning for SAR ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, Mar. 2022.

[37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[38] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[39] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[40] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 13733–13742.

[41] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3490–3499.

[42] H. Zhang, Y. Wang, F. Dayoub, and N. Sunderhauf, "VarifocalNet: An IoU-aware dense object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8514–8523.

[43] X. Li, C. Lv, W. Wang, G. Li, L. Yang, and J. Yang, "Generalized focal loss: Towards efficient representation learning for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3139–3153, Mar. 2023.

[44] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[45] Z. Huang, L. Huang, Y. Gong, C. Huang, and X. Wang, "Mask scoring R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6409–6418.

[46] T. Vu, H. Kang, and C. D. Yoo, "SCNet: Training inference sample consistency for instance segmentation," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2021, vol. 35, no. 3, pp. 2701–2709.

[47] Z. Tian, C. Shen, and H. Chen, "Conditional convolutions for instance segmentation," in *Proc. 16th Eur. Conf.* Glasgow, U.K.: Springer, Aug. 2020, pp. 282–298.

[48] Z. Tian, C. Shen, X. Wang, and H. Chen, "BoxInst: High-performance instance segmentation with box annotations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5443–5452.

[49] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.

[50] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7363–7372.

[51] X. Zhang, F. Wan, C. Liu, R. Ji, and Q. Ye, "FreeAnchor: Learning to match anchors for visual object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–9.

[52] Y. Wu et al., "Rethinking classification and localization for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10186–10195.

[53] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[54] C. Chen, C. He, C. Hu, H. Pei, and L. Jiao, "A deep neural network based on an attention mechanism for SAR ship detection in multiscale and complex scenarios," *IEEE Access*, vol. 7, pp. 104848–104863, 2019.

[55] J. Jiao et al., "A densely connected end-to-end neural network for multiscale and multiscene SAR ship detection," *IEEE Access*, vol. 6, pp. 20881–20892, 2018.

[56] Y. Zhao, L. Zhao, B. Xiong, and G. Kuang, "Attention receptive pyramid network for ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2738–2756, 2020.

[57] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.

[58] S. Wei et al., "Precise and robust ship detection for high-resolution SAR imagery based on HR-SDNet," *Remote Sens.*, vol. 12, no. 1, p. 167, Jan. 2020.

[59] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2018.

[60] J. Fu, X. Sun, Z. Wang, and K. Fu, "An anchor-free method based on feature balancing and refinement network for multiscale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1331–1344, Feb. 2021.

[61] Y. Liu, H. Li, C. Hu, S. Luo, Y. Luo, and C. Wen Chen, "Learning to aggregate multi-scale context for instance segmentation in remote sensing images," 2021, *arXiv:2111.11057*.

[62] H. Su et al., "HQ-ISNet: High-quality instance segmentation for remote sensing imagery," *Remote Sens.*, vol. 12, no. 6, p. 989, Mar. 2020.

[63] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[64] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," 2019, *arXiv:1904.01355*.

[65] P. Sun et al., "Sparse R-CNN: End-to-end object detection with learnable proposals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14454–14463.

[66] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," 2022, *arXiv:2201.12329*.

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